

Solar Flare Effects on Satellites

CAPSTONE PROJECT ~ AKSHAR PATEL



Main Problem



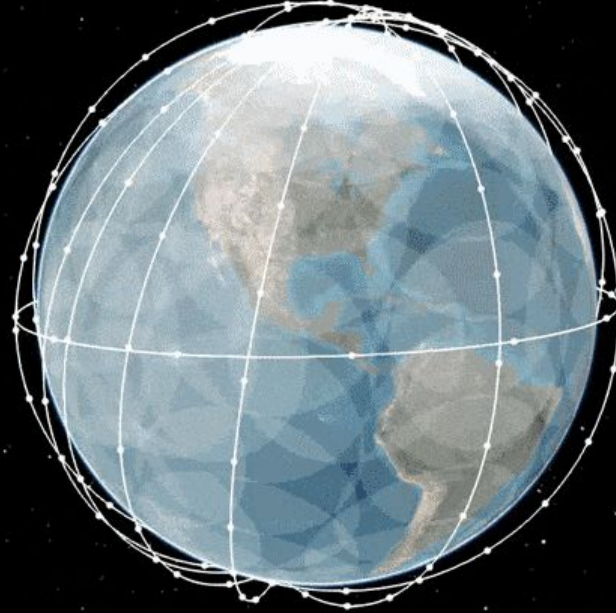
Geomagnetic Storms and Solar Flares have been prone to destroy data and corrupt communication.



Satellites help with GPS mapping, internet communications, weather forecasting, and telescopes for research purposes.



The reason why engineers haven't put more effort into resolving the issue is the cost of running through trials.



Attempted Solution



Considering that there are 7,702 satellites in active orbit and with projects such as Starlink, this innovation is crucial in building a global orbital network.



Solar flares contain high levels of energy; I will attempt to harness that and use it to power the satellites and implement a device to be used on the International Space Station.



Recent Storm



An intense geomagnetic storm hit Earth on February 10, 2024 which caused aurora borealis lights in uncommon areas.



Messed with GPS for tractors that required less than a centimeter of precession



"Satellites had a degraded service"
- Elon Musk on the Starlink System



Recent Damages + Sun Cycle



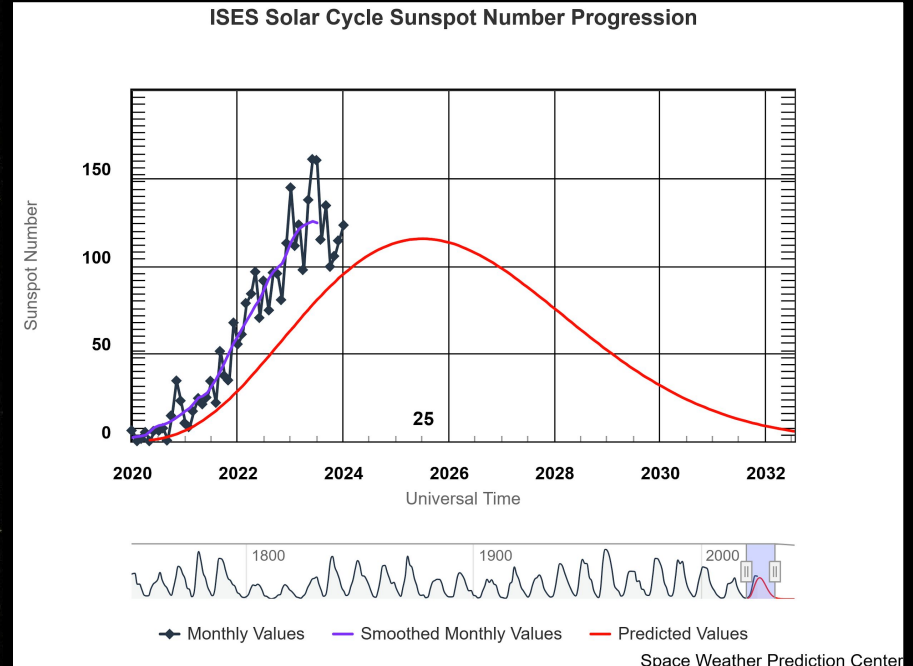
February 4, 2022 Starlink Launch



49 Satellites Destroyed = \$50 Million in Damages



Sunspots are the spots on the sun which Flares and geomagnetic fields can originate from



Starlink Fail



Destructive Interference



When two opposite waves collide and cancel each other.



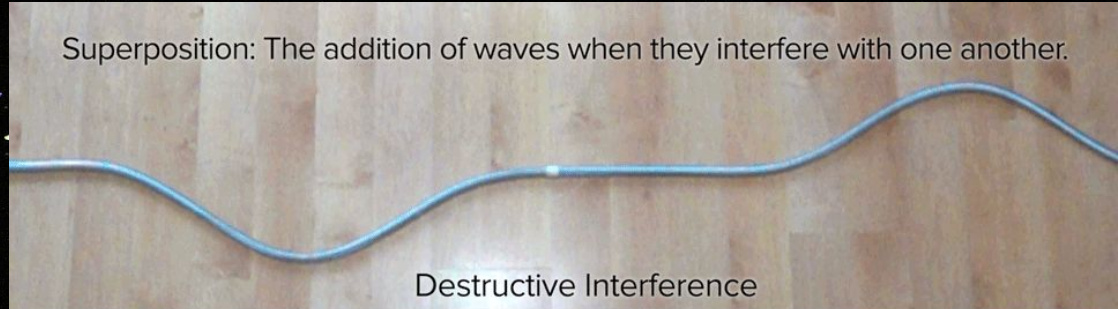
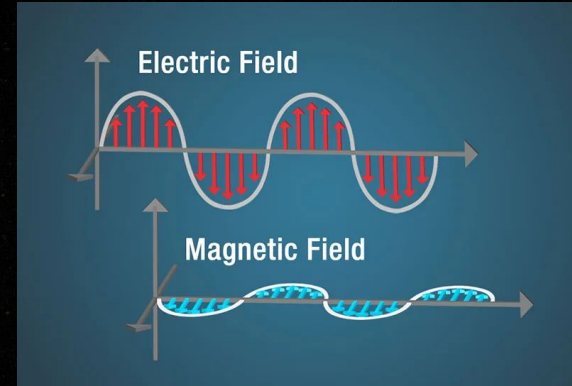
Generating electromagnetic waves to counteract storms



In this way we are able to protect communication and GPS networks



The types of rays emitted by geomagnetic storms and solar flares are x-rays, radio waves, and gamma rays.



Onshape Model



This solution is customizable depending on missions.



The process for these satellites are to detect the frequencies of the waves emitted by the sun and generating an opposite wave to counteract it



The 4 dishes serve for 360° coverage around the satellite are used to emit the counter waves

